

MIDTERM EXAM, TERM 2025.1 Course: Probability and Statistics. Course ID: MI2020E <i>Time duration: 40 minutes</i>		Full name: Student's ID: Numerical order: Class ID:
Full mark	Evaluator's name and signature	Proctor's name and signature

Exam number: 509 (The exam consists of 15 questions)

Remark: No document is allowed.

I. Single choice questions (Students need to find exactly one correct answer)

Câu 1. Let Y be a continuous random variable with the following probability density function $f(y) = \begin{cases} Cy & \text{if } 0 \leq y \leq 2 \\ 0 & \text{otherwise,} \end{cases}$ where C is a normalizing constant. Then $P(0.5 < Y < 1.5)$ equals to

- A. 0.5** **B. 0.75** **C. 0.375** **D. 0.625**

Câu 2. In a random experiment, let A, B be two events of a random experiment such that $P(B) = 0.45$ and $P(AB) = 0.2$. Then $P(A + \overline{B})$ equals to

- A. 0.55** **B. 0.25** **C. 0.65** **D. 0.75**

Câu 3. Let A, B, C be three events that are independent in pairs of a random experiment. Which of the following statements is incorrect?

- A.** $P(ABC) = P(A)P(B)P(C|AB)$ **C.** $P(ABC) = P(A)P(B)P(C)$
B. $P(AB) = P(A)P(B)$ **D.** $P(A + C) = P(A) + P(C) - P(A)P(C)$

Câu 4. The dropout rate among students in an online course is 15%. The course has 200 enrolled students. Let X be the number of students who drop out. Then $E(X^2)$ is

- A. 900** **B. 25.5** **C. 55.5** **D. 925.5**

Câu 5. The Rapid Test is used to determine whether someone has COVID-19 virus (Coronavirus Disease 2019). The test has a false positive rate of 2.5% and a false negative rate of 5%, respectively. In a certain region, 45% of the residents are infected with COVID-19 virus. A person is selected at random from this region and tested for COVID-19 virus. Given that the test result is positive, the probability that the person actually has the virus is

- A. 0.45** **B. 0.95** **C. 0.969** **D. 0.964**

Câu 6. At an airline ticket counter, an average of 6 customers arrive to buy tickets every 10 minutes. The probability that no more than one customer arrives within a 5-minute interval (rounded to three decimal places) is

- A. 0.149** **B. 0.287** **C. 0.199** **D. 0.205**

Câu 7. There are 5 candidates to be arranged into 3 exam rooms, numbered I, II, and III. How many ways are there to arrange them so that each room has at least one candidate?

- A. 240** **B. 150** **C. 210** **D. 160**

Câu 8. Let $\{A_1, A_2, A_3, A_4\}$ be a system of mutually exclusive and exhaustive events such that $P(A_1 + A_2) = 0.45$ and $P(A_2 + A_3 + A_4) = 0.9$. Then $P(A_2)$ equals to

- A. 0.1** **B. 0.25** **C. 0.3** **D. 0.35**

Câu 9. At a certain airport, two radar stations operate independently. The probabilities that they detect an arriving airplane are 0.65 and 0.75, respectively. The probability that the airplane is detected by exactly one of the two radar stations is

- A. 0.425 B. 0.4875 C. 0.575 D. 0.9125

Câu 10. A certain electrical system consists of 5 independent components. Each component has a probability of failure of 0.25. Given that at least one component has failed, the probability that exactly three components have failed is

- A. 0.2373 B. 0.1153 C. 0.0879 D. 0.7627

II. Multiple choice questions (Students need to find ALL correct answers)

Câu 11. A multiple-choice examination contains 15 questions, each offering four possible answers, with only one correct option per question. Each correct answer is awarded 2 points, while incorrect answers receive 0 points. A student completes all 15 questions by randomly selecting one option for each question. Let X be the total score obtained by the student. Which of the following statements is correct?

- A. $E(X) = 3$. D. $P(X \geq 6) = 0.7639$
 B. $P(X \geq 6) = 0.6793$ E. $P(X = 10) = 0.1651$
 C. $P(X = 10) = 0.1561$ F. $E(X) = 4$

Câu 12. The discrete random variable X has the following probability distribution

X	1	3	5	7	9
$P(X = x_i)$	0.2	c	0.2	0.3	0.1

Let $Y = \max\{X, 4\}$. Which of the following statements are correct?

- A. $V[Y] = 3.02$ D. $E[Y] = 5.6$
 B. $E[Y] = 5.25$ E. $E[X] = 5.5$
 C. $V[Y] = 2.84$ F. $\text{Mod}[Y] = 7$

Câu 13. The length of time in minutes that a customer queues in a bank is a random variable T with PDF $f(t) = C(36 - t^2)$ if $t \in [0, 6]$, and $f(t) = 0$ if $t \notin [0, 6]$, where C is a normalizing constant. Which of the following statements are correct?

- A. $P[T > 3] = 0.4125$ C. $P[T > 3] = 0.3875$ E. $E[T^2] = 7.6$
 B. $E[T^2] = 7.5$ D. $P[T > 3] = 0.3125$ F. $E[T^2] = 7.2$

Câu 14. Researchers have developed statistical models based on financial ratios that predict whether a company will go bankrupt over the next 12 months. In a test of one such model, the model correctly predicted the bankruptcy of 80% of firms that did in fact fail, and it correctly predicted nonbankruptcy for 70% of firms that did not fail. Suppose that we expect 10% of the firms in a particular city to fail over the next year. Let A be the event that "A randomly chosen firm will actually go bankrupt within the next 12 months" and let B be the event that "The model will predict bankruptcy for a randomly chosen firm".

- A. $P(B) = 0.2$ D. $P(B) = 0.35$
 B. $P(B) = 0.3$
 C. $P(A|B) = 0.292$ E. $P(A|B) = 0.229$

Câu 15. In a random experiment, let A and B be two independent events. Which of the following statements are correct?

- A. $P(A) + P(B) = 1$ C. $P(A\bar{B}) = P(A)[1 - P(B)]$ E. $P(AB) = 0$
 B. $P(AB) = P(A)P(B)$ D. $P(A + B) = P(A) + P(B)$ F. $P(A\bar{B}) = P(A) - P(AB)$